

Programme	BSc Mathematics Honours			
Course Code	MAT3VN202			
Course Title	DATA MINING			
Type of Course	Vocational Minor – Introduction to Data Science			
Semester	III			
Academic Level	200-299			
Course Details	Credit	Lecture/Tutorial	Practical	Total Hours
		per week	per week	
	4	3	2	75
Pre-requisites	Basic Knowledge in MS Excel			
Course Summary	Course aims to provide basic data mining techniques using Weka tool			

Course Outcome:

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the fundamental concepts and principles of data mining	U	C	Internal exam/ Assignment/ Seminar/ External/ Practical Assessment
CO2	Understand the mining techniques like association, classifications and clustering on datasets	U	P	Internal exam/ Assignment/ Seminar/ External/ Practical Assessment
CO3	Apply data mining techniques to real-world datasets	Ap	P	Internal exam/ Assignment/ Seminar/ External/ Practical Assessment
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Ext. Marks (70)
I		Introduction to Data Mining	8	Min 15
	1	Data Warehousing - Data warehousing architecture, Warehouse Schema, Data warehouse backend process, Multidimensional Data Model		
	2	OLAP Operations, Introduction to KDD process, Data mining		
	3	Data mining Functionalities, Classification of Data Mining Systems.		
	4	Data Warehousing Case Study: Government, Tourism and Industry		
	5	Data Preprocessing - Data Cleaning, Data Integration and Transformation, Data Reduction, Data discretization		
II		Association Analysis	7	Min 15
	6	Association Analysis - Basic Concepts, Frequent Item set Mining Methods: Apriori Algorithm, generating association Rules from Frequent Item sets, Improving the Efficiency of Apriori.		
	7	Evaluation of Association Patterns, Visualization, Partition algorithm A Case Study on Association using Orange Tool		
	8	Dynamic Item set Counting algorithm- FP-tree growth algorithm-Incremental Algorithm-Border algorithm		
III		Classification & Prediction	14	Min 15
	9	Classification Technique: Introduction, Decision Trees: Tree Construction Principle – Attribute Selection measure – Tree Pruning - Decision Tree construction Algorithm – CART – ID3		
	10	Bayesian Classification: Bayes’ theorem, Naïve Bayesian Classification		
	11	K- Nearest Neighbour Classifiers, Support Vector Machine. Evaluating the performance of a Classifier, Methods for comparing classifiers, Visualization		
	12	Case Study of Classification using Orange Tool		

	13	Linear Regression, Nonlinear Regression, Other Regression-Based Methods		
IV		Clustering	16	Min 15
	14	Clustering techniques: Data Attribute Types – Data Similarity and Dissimilarity		
	15	Partitioning Methods: k-Means and k-Medoids, CLARANS		
	16	Hierarchical Method: Agglomerative and Divisive Hierarchical Clustering		
	17	Density-based Clustering - DBSCAN, Grid based clustering-STING		
	18	Evaluation of Clustering Method		
	19	Case Study of Clustering using Orange Tool		
	20	Introduction to Web Mining - Basic concepts, Web content mining, Web structure mining, Web usage mining		
	21	Introduction to Text mining, Text Preprocessing, Text clustering		
	22	Case Study – Web Mining: Analysing User Behaviour on E-commerce Website Case Study - Sentiment Analysis of Customer Reviews		
		Practical's	30	
	1	Installation of WEKA Tool		
	2	Creating new Arff File		
	3	Pre-Processes Techniques on Data Set		
	4	Pre-process a given dataset based on Handling Missing Values		
	5	Generate Association Rules using the Apriori Algorithm		
	6	Generating association rules using FP growth algorithm		
	7	Build a Decision Tree by using ID3 algorithm		
	8	Build a Naïve Bayesian Classifier		
	9	Build a K- Nearest Neighbour Classifiers		
	10	Build a Support Vector Machine		

	11	Build a Linear Regression		
	12	Build K-Means Algorithm		
	13	Build K-Medoids Algorithm		
	14	Build Hierarchical Clustering Algorithms		
	15	Create Student. ariff file to suggest better college using Decision tree		
		References		
	1	Arun K Pujari, "Data Mining Techniques", Universities Press. 2012		
	2	Pang-Ning Tan, Michael Steinbach, Vipin Kumar, 'Introduction to Data Mining'		
	3	G. K. Gupta, "Introduction to Data Mining with Case Studies", Easter Economy Edition, Prentice Hall of India, 2006.		
	4	Data Mining: Practical Machine Learning Tools and Techniques" by Ian H. Witten, Eibe Frank, Mark A. Hall, and Christopher J. Pal:		
	5	Data Mining: Concepts and Techniques" by Jiawei Han, Micheline Kamber, and Jian Pei:		

Note: Proofs of all the results are exempted for the end semester exam.

Mapping of COs with PSOs and POs :

	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	2	3	2	3	2	3	1	2
CO 2	2	3	3	2	3	2	3	1	2
CO 3	2	3	3	2	3	2	3	1	2

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓